Automation without Borders: The Impacts of Generative AI on International Service Outsourcing

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Any opinions and conclusions expressed in this discussion do not necessarily represent the views of the Banque de France.

This paper: What it does

- Provides empirical evidence on how generative AI is reshaping international outsourcing in digital services
- Exploits rich, high-frequency microdata from a leading Online Labor Marketplace covering the period 2022-2024
- Develops skill-level exposure measures based on ChatGPT's self-assessed ability to assist/complement or substitute for each skill
- Estimates the effects of AI exposure on job volumes, job value, worker earnings, and the reallocation of tasks across skills and countries

This paper: Why is it important?

- International service outsourcing has become a key growth engine for many developing economies (e.g., India)
- Earlier waves of technological change (e.g., robotization) had geographically bounded effects, whereas GenAI represents a form of automation that is largely unbounded by geography
- The paper speaks directly to major debates on:
  - whether GenAI substitutes for or complements global digital labor
  - whether it compresses wages
  - how workers and firms reorganize tasks and production processes in response to AI
- The topic is central for policymakers concerned with cross-country inequality and the future of global digital trade

This paper: What it finds

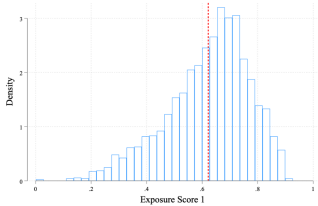
- Decline in outsourcing: at mean exposure, international job volumes fall by about 25%, and job value by roughly 35%. But continuing workers earn more.
- Effects are strongly concentrated in developing countries and in skill areas with high AI substitutability
- Firms post fewer tasks overall but shift their demand toward higher-value, more complex projects
- Substantial reskilling and task reallocation: workers transition toward AI-complementary or adjacent skills

General Comment

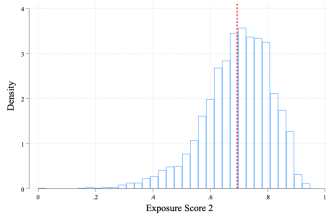
- Timely paper addressing a major question at the intersection of AI, labor markets, and international trade
- It leverages rich and, to my knowledge, original microdata from a global Online Labor Marketplace
- The results point to a significant reconfiguration of global task flows, with potentially sizable losses for developing economies
- The paper makes a valuable and policy-relevant contribution, although the interpretation of the underlying mechanisms could be further clarified

Comment #1: Measuring AI exposure

Distribution of Score 1 (Assist)

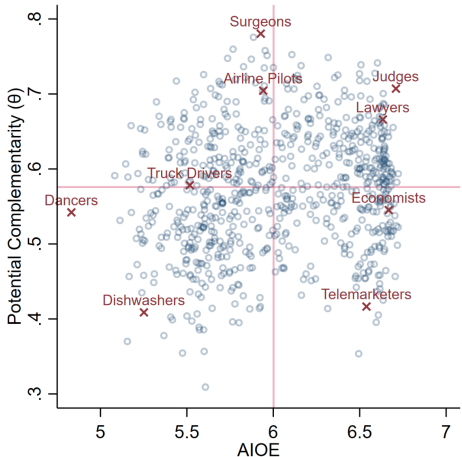


Distribution of Score 2 (Substitute)



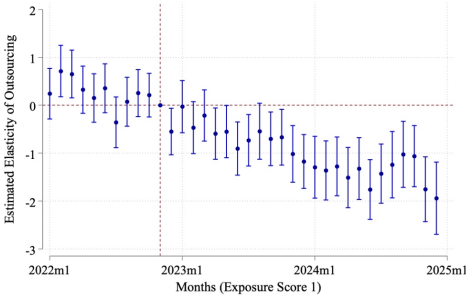
- The paper relies on ChatGPT's self-assessed ability to assist or substitute for human skills
- Exposure scores are very high on average (around 7) and strongly correlated (0.9) across the two measures (complementarity and substituability)
  - Some skill categories are extremely broad (e.g., "Software Engineering", "Writing"), which may partly explain why ChatGPT systematically reports high scores
  - The current approach may not fully distinguish between skills for which AI is likely to be complementary and those for which it is likely to be substitutable

Comment #1: Measuring AI exposure



- **Suggestion:** provide additional validation using established external benchmarks (e.g., ILO (2023) using O*NET tasks; IMF (2023) using O*NET abilities), which could help better differentiate between complementarity and substitutability channels

Comment #2: Empirical Approach

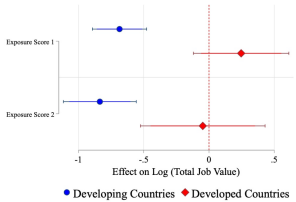


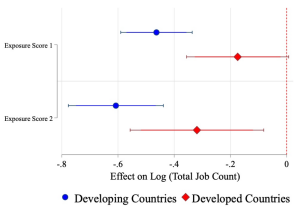
- The before/after diff-in-diff design does not explicitly document whether pre-treatment trends are parallel across skills with different levels of AI exposure.
- The event-study specification (dynamic diff-in-diff?) strengthens the credibility of the results, though pre-trends may remain a potential concern

Comment #2: Empirical Approach

- Endogeneity concern: evolution of skill content
  - The emergence of GenAI may itself change the underlying content of many skills
  - If the skills that appear most "AI-exposed" are also those undergoing the fastest transformation then the estimated decline in activity may partly capture a temporary mismatch: firms demand updated skills while workers have not yet fully adapted or acquired the required training
- IV strategy to further isolate the causal effect?

Comment #3: Interpretation of Results and Mechanisms





- The negative impact on job value appears bigger than on job count in developing countries
- By contrast, in developed countries, job counts decline but job value do not fall
- The increase in the average value of remaining jobs seems to be driven primarily by developed countries rather than developing ones
- How should we interpret this asymmetric pattern? Does it reflect task upgrading, selection of workers?